

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from scipy import stats
```

```
np.random.seed(42)
n = 200

TV = np.random.uniform(0.1, 300, n)
Radio = np.random.uniform(0.1, 50, n)
Social_Media = np.random.uniform(0.1, 30, n)
Sales = 2.5 + 0.045*TV + 0.19*Radio + 0.12*Socia_Media + np.random.normal(0,

df = pd.DataFrame({'TV': TV, 'Radio': Radio, 'Social_Media': Social_Media, 'Sa
print(df.shape)
df.head()
```

(200, 4)

	TV	Radio	Social_Media	Sales
0	112.424582	32.137379	3.183404	13.251830
1	285.219220	4.298584	27.086332	18.212647
2	219.624983	8.165273	15.207046	15.598826
3	179.637679	44.937854	24.811078	20.546354
4	46.889990	30.360810	9.669483	10.708467

Double-click (or enter) to edit

```
print(df.describe())
print("\nMissing values:\n", df.isnull().sum())

sns.pairplot(df)
plt.suptitle("Pairplot of Marketing Variables", y=1.02)
plt.show()

plt.figure(figsize=(8,5))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Matrix")
plt.show()
```

	TV	Radio	Social_Media	Sales
count	200.000000	200.000000	200.000000	200.000000
mean	145.253469	25.268319	15.668116	15.628316
std	88.437940	14.620759	9.188533	4.977823
min	1.756083	0.352573	0.424046	2.366087
25%	68.651866	13.146979	7.740057	11.708574
50%	148.396427	27.127830	15.809444	15.834320
75%	227.082199	37.135746	24.352335	19.284070
max	296.067392	49.526207	29.991558	25.639877

Missing values:

TV 0

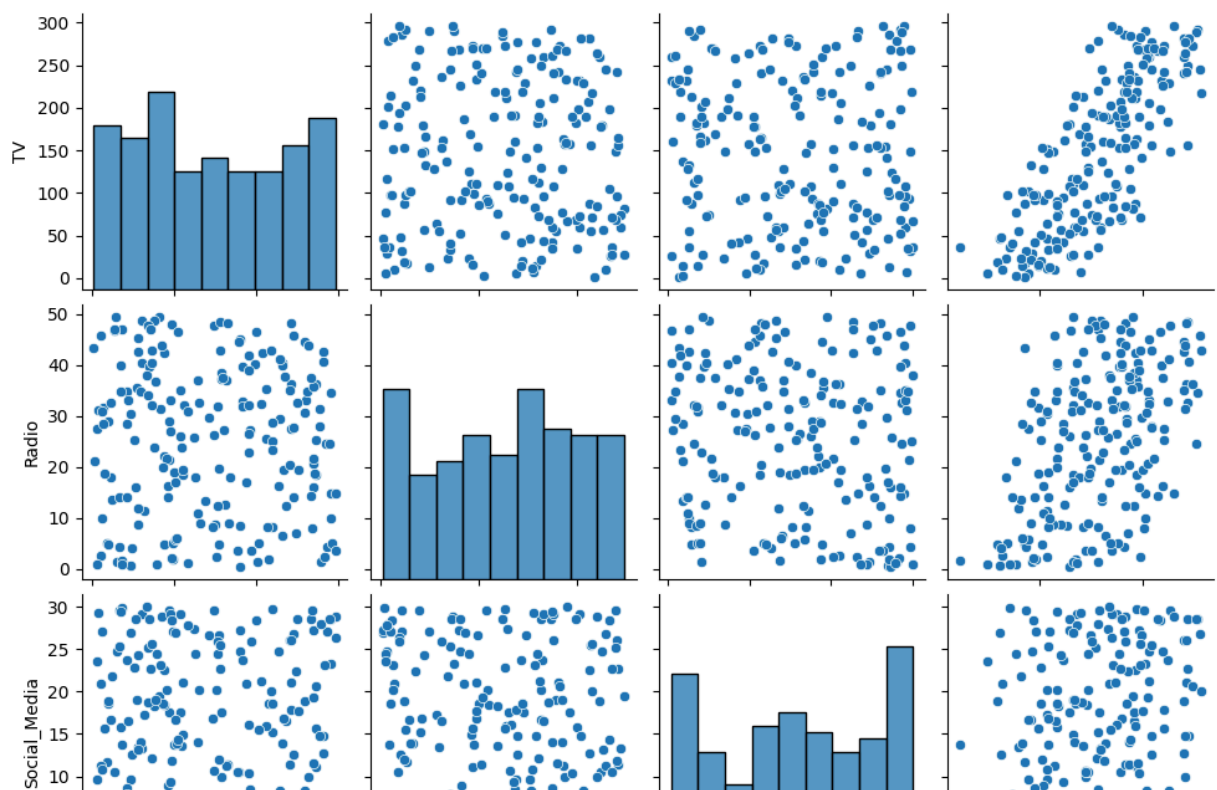
Radio 0

Social_Media 0

Sales 0

dtype: int64

Pairplot of Marketing Variables



```

correlations = df.corr()['Sales'].drop('Sales')
print("Correlations with Sales:\n", correlations)
best_predictor = correlations.idxmax()
print(f"\nBest predictor: {best_predictor}")

```

Correlations with Sales:

TV 0.769659

Radio 0.483763

Social_Media 0.144257

Name: Sales, dtype: float64

Best predictor: TV

```
X = df['TV']
y = df['Sales']

X_const = sm.add_constant(X)
model = sm.OLS(y, X_const).fit()
print(model.summary())
```

OLS Regression Results

```
=====
Dep. Variable:          Sales    R-squared:
Model:                  OLS      Adj. R-squared:
Method:                 Least Squares    F-statistic:
Date:                  Tue, 09 Jun 2026    Prob (F-statistic):      1.9
Time:                  10:14:41           Log-Likelihood:        -5
No. Observations:      200             AIC:
Df Residuals:          198             BIC:
Df Model:               1
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0
const	9.3358	0.434	21.510	0.000	8.480	1
TV	0.0433	0.003	16.963	0.000	0.038	

```
=====
Omnibus:                2.072    Durbin-Watson:
Prob(Omnibus):          0.355    Jarque-Bera (JB):
Skew:                   0.029    Prob(JB):
Kurtosis:               2.560    Cond. No.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is co

```
fitted = model.fittedvalues
residuals = model.resid

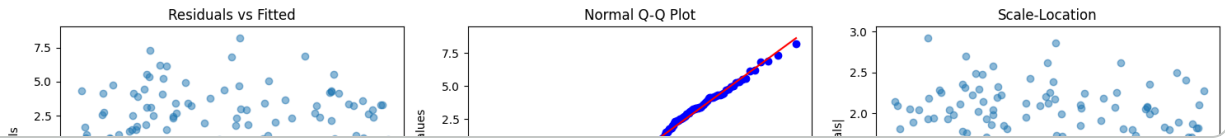
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

# Linearity
axes[0].scatter(fitted, residuals, alpha=0.5)
axes[0].axhline(0, color='red', linestyle='--')
axes[0].set_title("Residuals vs Fitted")
axes[0].set_xlabel("Fitted Values")
axes[0].set_ylabel("Residuals")

# Normality
stats.probplot(residuals, dist="norm", plot=axes[1])
axes[1].set_title("Normal Q-Q Plot")

# Homoscedasticity
axes[2].scatter(fitted, np.sqrt(np.abs(residuals)), alpha=0.5)
axes[2].set_title("Scale-Location")
axes[2].set_xlabel("Fitted Values")
axes[2].set_ylabel("√|Residuals|")
```

```
plt.tight_layout()
plt.show()
```



```
r2 = model.rsquared
coef_TV = model.params['TV']
pval_TV = model.pvalues['TV']

print("=" * 50)
print("MODEL INTERPRETATION")
print("=" * 50)
print(f"R-squared: {r2:.4f}")
print(f"TV Coefficient: {coef_TV:.4f}")
print(f"P-value (TV): {pval_TV:.6f}")

print("""
BUSINESS RECOMMENDATION:
- TV advertising has the strongest correlation with Sales.
- Every $1 increase in TV budget increases Sales by {:.4f} units.
- R2 of {:.2%} means TV explains {:.2%} of Sales variance.
- P-value < 0.05 confirms TV is statistically significant.
- Recommendation: Prioritize TV budget allocation for maximum ROI.
""").format(coef_TV, r2, r2))
```

```
=====
MODEL INTERPRETATION
=====
```

```
R-squared: 0.5924
TV Coefficient: 0.0433
P-value (TV): 0.000000
```

```
BUSINESS RECOMMENDATION:
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- TV advertising has the strongest correlation with Sales.
- Every \$1 increase in TV budget increases Sales by 0.0433 units.
- R² of 59.24% means TV explains 59.24% of Sales variance.
- P-value < 0.05 confirms TV is statistically significant.
- Recommendation: Prioritize TV budget allocation for maximum ROI.

